5.2 Classification

Carbohydrates are often referred to as saccharides. They are classified into three main groups

i) Monosaccharides ii) Oligosaccharides iii) Polysaccharides.

5.2.1 Monosaccharides:

Monosaccharides are the simplest group of carbohydrates which cannot be hydrolysed further. They are often referred to as "simple sugars". They have the general formula $C_n(H_2O)_n$. They can be further subdivided on the basis of number of carbon atoms (triose, tetrose, pentose etc.) and on the basis of the functional group (aldose, ketose) as given in Table – 5.1.

Table – 5.1 Classification of monosaccharides

Number of 'C' atoms	Name of the monosaccharide	Aldose	Ketose
3	Triose	Glyceraldehyde (Glycerose)	Dihydroxyacetone
4	Tetrose	Erythrose	Erythrulose
5	Pentose	Ribose	Ribulose
6	Hexose	Glucose	Fructose
7	Heptose	Glucoheptose	Sedoheptulose

5.2.2 Oligosaccharides :

Oligosaccharides are the carbohydrates which yield two to ten monosaccharide units on hydrolysis. They are further classified into di, tri, tetra (or) penta saccharides respectively containing 2, 3, 4 (or) 5 monosaccharide units.

Example:

a) Maltose (disaccharide) (Glucose + Glucose)

b) Sucrose (disaccharide) (Glucose + Fructose)

c) Lactose (disaccharide) (Glucose + Galactose)

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